

Explainable Machine Learning for Pre-Symptomatic Crop Disease Risk Prediction from IoT Micro-Climate Data

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Abstract—Crop losses caused by pathogens are strongly governed by micro-climate, yet the tools available to small-scale and urban growers remain either symptom-dependent or infrastructure-heavy. Image-based classifiers cannot act until a lesion is visible, and greenhouse forecasting systems typically assume farm-grade instrumentation and offer little insight into why a warning was issued. This paper presents an explainable, sensor-only machine-learning framework that estimates crop-disease risk directly from consumer-grade IoT micro-climate readings—temperature, humidity, rainfall, soil moisture, soil pH and light intensity—together with crop type and season. A working set of 1,652 IoT environmental records covering potato and tomato was labelled across six outcome classes (Healthy plus five diseases) using condition thresholds taken from published plant-pathology extension references. Random Forest and XGBoost classifiers were trained on an 80:20 stratified split with balanced class weighting and compared under repeated stratified 5-fold cross-validation (25 runs). XGBoost was the stronger model on every aggregate metric, reaching 95.17% accuracy and a macro F1-score of 0.9132 on the 331-sample held-out test set against 92.45% and 0.8584 for Random Forest; a Wilcoxon signed-rank test over the 25 paired validation scores confirmed the gap is statistically significant ($W = 12.0$, $p < 0.001$). SHAP analysis of the selected model ranked soil moisture, crop type and temperature as the dominant contributors, agreeing closely with Random Forest impurity importance (Spearman $\rho = 0.8909$) while contradicting XGBoost gain importance ($\rho = -0.2848$), which shows that single-method importance reporting is unsafe in this setting. Minority classes remained the hardest to detect, with Common Scab the weakest class for both models. The results indicate that interpretable, pre-symptomatic disease-risk screening is attainable from low-cost sensor data alone, while also exposing the overfitting and label-provenance limits that must be addressed before field deployment.

Keywords—crop disease prediction, Internet of Things, machine learning, Random Forest, XGBoost, explainable AI, SHAP, precision agriculture, urban farming, micro-climate

I. INTRODUCTION

Plant disease development is, to a large degree, an environmental process. The infection of a susceptible host requires not only the presence of a pathogen but a window of favourable conditions warmth, surface wetness, soil moisture and a suitable soil reaction sustained for long enough to permit germination and colonisation. This dependence is the reason weather-driven advisory systems have been in agricultural use for decades, and it is also the reason climate

variability is expected to reshape where and when particular diseases appear [19], [20].

The computational response to this problem has developed along two largely separate lines. The first is image-based classification, in which convolutional and transformer architectures assign a disease label to a photograph of a leaf. Accuracy on curated benchmarks is high 98.66% for the PLDPNet hybrid segmentation-and-classification pipeline on potato leaves [10], and 98.17% for a fine-tuned enhanced CNN deployed as a mobile application across apple, corn, and potato [11]. These systems are, however, structurally reactive: a visible lesion is a precondition for a prediction, so the diagnosis arrives after infection has already succeeded. The second line is environmental forecasting, in which weather or greenhouse sensor streams are mapped onto a risk score. Representative work includes an autoencoder trained on 48-hour rolling environmental statistics that achieved a mean AUROC of 0.917 across several crops [1], [8], a support vector machine for rice blast reaching $r = 0.77$ [9], and an XGBoost, LSTM and Random Forest comparison for greenhouse micro-climate in tomato production in which XGBoost was most accurate, with R^2 up to 0.95 [4]. These systems can warn before symptoms emerge, but they typically presuppose greenhouse-grade or research-grade instrumentation, and—as review studies have repeatedly noted [6], [14] they seldom explain their outputs.

Neither line addresses the grower who is arguably most exposed to the consequences of a missed diagnosis: the urban, balcony or smallholder cultivator, who has no agronomist on call, no leaf-wetness sensor, and no realistic path to greenhouse instrumentation, but who may well own a handful of inexpensive temperature, humidity, soil-moisture and light sensors. For that user, a prediction is only useful if it arrives before symptoms and if it can be acted upon, which in practice means the system must be able to say which measured condition drove the warning.

This paper addresses that gap. The contributions are as follows.

- A sensor-only disease-risk classifier is constructed over six low-cost micro-climate variables plus crop type and season, with no imagery and no specialised agricultural instrumentation required at inference time.
- Random Forest and XGBoost are benchmarked under an identical imbalance-aware protocol and compared not on a single split but over 25 repeated

stratified cross-validation runs, with the difference subjected to a Wilcoxon signed-rank test.

- Performance is reported per disease class rather than in aggregate only, so that the behaviour of the framework on the minority classes—precisely the classes an early-warning system exists to catch—is made explicit.
- SHAP is used at global, class-specific and individual-prediction levels, and the resulting rankings are cross-checked against both models’ built-in importance measures using Spearman rank correlation. This cross-check reveals a substantive disagreement between gain-based importance and Shapley attribution that would have produced a misleading agronomic conclusion had a single method been trusted.
- The limitations arising from rule-derived labelling and from the observed train-validation gap are reported explicitly rather than omitted, so that the strength of the evidence is not overstated.

The remainder of the paper is organised as follows. Section II reviews related work and identifies the research gap. Section III describes the dataset and the labelling procedure. Section IV sets out the methodology. Section V reports results. Section VI discusses their interpretation, Section VII states the limitations, and Section VIII concludes.

II. RELATED WORK

Prior work on computational crop-disease prediction can be grouped into four strands: image-based classification, environmental and weather-driven forecasting, hybrid and review studies, and the pre-machine-learning statistical models that preceded all of them. Table I summarises the studies most directly comparable to the present work.

TABLE I. SUMMARY OF REPRESENTATIVE RELATED WORK

Ref.	Focus	Model	Reported best result
[1], [8]	Crop pest/disease risk from sequential environmental data	Linearization autoencoder	Mean AUROC 0.917; 0.8755 on independent validation
[2]	Cucumber downy mildew early warning in greenhouses	Greenhouse energy-balance model + infection model	72-hour advance warning
[3]	Plant disease classification and detection (review)	YOLOv5, DenseNet121, ResNet50, MobileNetV2	DenseNet121 best classification accuracy, 61.01%
[4]	Greenhouse micro-climate forecasting in tomato	XGBoost, LSTM, Random Forest	XGBoost best; R^2 up to 0.95 (soil temperature)
[7]	Apple scab risk zoning from IoT sensors	Multiple linear regression	Safe/unsafe orchard zoning across three regions
[9]	Rice blast prediction from weather data	SVM vs. regression, BPNN, GRNN	SVM best; $r = 0.77$, %MAE = 36.66%
[10]	Potato leaf disease prediction	U-Net + VGG19 + Inception-V3 + ViT hybrid	98.66% accuracy
[11]	Apple/corn/potato fungal disease detection	Enhanced CNN with mobile app	98.17% accuracy
[12]	Pre-symptomatic potato Early Blight and Gray Mold	RF, LR, KNN, DT, XGBoost	Group model 90%; RF 82% (Early Blight), XGBoost 94% (Gray Mold)
[14]	Plant and crop disease forecasting (review of 46 studies)	SVM, ANN, RF, LSTM, CNN and others	Weather-based models dominate (63% of studies)
[15]	Greenhouse micro-climate prediction from fixed sensors	Dense neural network	RMSE reduced by 68.67% (temperature), 46.21% (RH)
This work	Six-class crop-disease risk from low-cost IoT micro-climate	Random Forest, XGBoost, SHAP	XGBoost 95.17% accuracy, macro F1 = 0.9132

A. Image-Based Detection

Convolutional and transformer models classify disease from leaf photographs, and purpose-built pipelines perform strongly on curated data: PLDPNet combined U-Net segmentation, ensembled VGG19 and Inception-V3 feature extraction and a Vision Transformer classifier to reach 98.66% accuracy on potato leaves [10], while an enhanced CNN packaged as a mobile application reached 98.17% across apple, corn and potato [11]. A comparative review of five detection algorithms and eighteen classification architectures found YOLOv5 strongest for detection and DenseNet121 strongest for classification, although its reported classification accuracy of 61.01% on realistic field imagery is far below the figures obtained under controlled conditions [3]. The common structural constraint is that all of these systems require a visible lesion, so they offer no pre-symptomatic lead time and make no use of the environmental drivers that precede infection.

B. Environmental and Weather-Based Forecasting

The second strand predicts risk from environmental measurements. An autoencoder with a linearised latent space converted 48-hour rolling statistics into a continuous risk score, achieving a mean AUROC of 0.917 and 0.8755 on an independent validation set [1], [8]. For rice blast, a support vector machine outperformed multiple regression and two neural baselines, reaching $r = 0.77$ with a percentage mean absolute error of 36.66% [9]. In greenhouse tomato production, XGBoost, LSTM and Random Forest were compared for micro-climate forecasting, with XGBoost most accurate at R^2 up to 0.95 for soil temperature and approximately 61% lower mean error than Random Forest [4]. Closest in spirit to the present study is the pre-symptomatic potato work of [12], which combined spore-trap counts with weather variables and found Random Forest and XGBoost jointly reaching 90% accuracy in a group model, with Random Forest best for Early Blight (82%) and XGBoost best for Gray Mold (94%). The consistency with which gradient boosting and Random Forest appear at the top of these comparisons motivated the choice of the same two families here.

C. Hybrid and Review Studies

Hybrid work links physical process models to forecasts: a one-dimensional transient greenhouse climate model was coupled to an infection-and-incubation model to give 72-hour advance warning of cucumber downy mildew [2]. Reviews provide the broader context. Of 46 studies surveyed in [14], 63% were weather-data-based, 22% image-based and 15% based on heterogeneous data, and the review noted that SVM and ANN models are habitually treated as black boxes. An integrative review of IoT, machine learning and AI for disease forecasting under climate change identified poor interpretability, scarce open data and infrastructure cost as the principal barriers to adoption in smallholder settings [6]. Sensor-network studies with a small number of variables have also been reported, including apple-scab zoning from temperature and leaf-wetness hours [7] and greenhouse micro-climate prediction from fixed sensor placements [15].

D. Pre-Machine-Learning Statistical Models

The variables used in modern models were largely established by earlier statistical work. Logistic regression was used to model rose downy mildew infection as a function of temperature and leaf-wetness duration, establishing optimal infection temperatures of 15–20 °C and showing that two hours of wetness could suffice at optimal temperature [24]. Field evaluation of the BOTCAST and SIV advisory models for Botrytis leaf blight in onion reduced fungicide applications by 54% relative to a weekly spray programme with no yield penalty [25]. These studies fixed the core predictor set—temperature, humidity and moisture duration—that machine-learning models now recombine with interactions.

E. Research Gap

Three gaps follow from this survey. Image-based models [3], [10], [11] require visible symptoms and therefore cannot serve as early warning. Environmental models [1], [4], [8], [9], [12] achieve pre-symptomatic prediction but are generally built for a single farm, crop or greenhouse and assume instrumentation beyond the reach of an urban grower. Across both strands, explainability is rarely reported: reviews [6], [14] identify interpretability as an open problem rather than a solved one, and even studies that acknowledge it frequently propose XAI as future work [10]. The present study targets the intersection of these three gaps by restricting the feature set to variables measurable with low-cost consumer sensors, by benchmarking two tree ensembles per disease class rather than in aggregate only, and by making Shapley-based explanation a reported result subject to cross-method validation rather than an afterthought.

III. DATASET AND DATA COLLECTION

The environmental measurements used in this study are drawn from the Bangladesh IoT Environmental Dataset (BIED), published on Mendeley Data [27]. BIED captures the data of various IoT sensors such as temperature, humidity, soil moisture, rainfall, soil pH, light intensity at urban, rural, coastal, agricultural and industrial areas of Bangladesh. The dataset carries no disease annotation, so a separate labelling stage was required.

A. Rule-Based Disease Labelling

Disease labels were assigned by comparing each record’s environmental profile against disease-conducive conditions documented in plant-pathology extension references and recording the closest matching disease class; records that matched no disease profile were labelled Healthy. The reference conditions were taken from UConn IPM for tomato Verticillium wilt [28], UC ANR IPM for potato and tomato Fusarium wilt [29], [30], the Cornell Vegetables Program for potato Early Blight and tuber defects [31], and NDSU Extension for potato Common Scab, which is associated with higher soil pH and drier soil [32]. Where a reference did not specify every variable—as for tomato Anthracnose and for Early Blight across both crops—the general condition of warm temperature with elevated humidity and moisture was applied. No sensor value was altered at any point in this stage; the procedure added the *disease_label* column and nothing else. The consequences of this labelling choice for the validity of the results are addressed directly in Section VII.

B. Composition of the Working Set

After cleaning, the working set comprises 1,652 observations described by eight raw attributes: six numerical environmental variables and two categorical variables (crop type and season). It contains 2 seasons (summer=1235, winter=417) and 2 crops (potato=874, tomato=778). These two categorical features are one-hot encoded to generate the 10-dimensional feature matrix to train. The target has six classes. Table II reports the class distribution and Fig. 1 displays it graphically.

TABLE II. DISEASE-LABEL CLASS DISTRIBUTION

Disease class	Count	Share (%)
Healthy	809	48.97
Verticillium Wilt	304	18.40
Fusarium Wilt	257	15.56
Early Blight	103	6.23
Anthracnose	91	5.51
Common Scab	88	5.33
Total	1,652	100.00

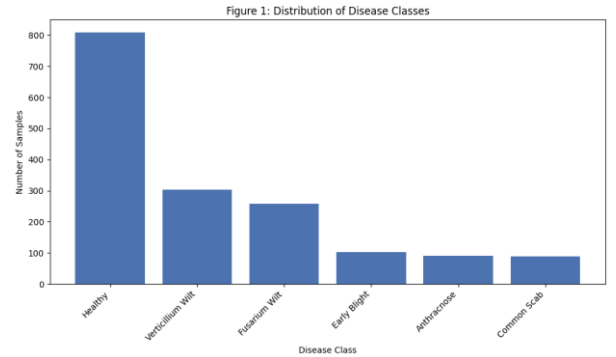


Fig. 1. The distribution of six classes of disease in the 1,652-record working set. Healthy constitutes 48.97% of the records; five disease classes form 51.03%.

The Healthy class accounts for 48.97% of records and the five disease classes together for 51.03%. The imbalance is not between healthy and diseased but among the disease classes themselves: Healthy contains 809 records against 88 for Common Scab, an imbalance ratio of approximately 9.19:1. This ratio, rather than the headline healthy/diseased balance, governs the evaluation strategy adopted in Section IV.

Table III reports summary statistics for the six numerical variables. Two distributional features are worth noting because they affect interpretation later. Rainfall is zero-inflated: its first, second and third quartiles are all 0.00 mm, so an interquartile-range test flags 334 records (20.22%) as outliers even though these are legitimate rainfall events rather than sensor faults. Light intensity is strongly bimodal, with a median of 99 lux against a mean of 12,524 lux, reflecting the mixture of night-time and daylight readings in the source data. Neither variable was trimmed or transformed, since both patterns are properties of the physical quantity rather than measurement error.

TABLE III. SUMMARY STATISTICS OF NUMERICAL FEATURES (N = 1,652)

Feature	Mean	Std.	Min.	Median	Max.
Temperature (°C)	29.82	5.28	15.47	31.32	42.00
Humidity (%)	67.14	7.96	42.00	67.48	88.99
Rainfall (mm)	2.08	6.38	0.00	0.00	50.00
Soil moisture (%)	42.22	14.11	10.00	41.30	93.15
Soil pH	6.17	0.48	5.50	6.14	7.58
Light intensity (lux)	12,524.37	15,338.30	0.00	99.00	50,000.00

Table IV reports the mean environmental profile of each class and provides a check on the plausibility of the labelling. The separations are consistent with the reference conditions used: Fusarium Wilt records have the lowest mean soil moisture (27.48%) and the highest mean soil pH (6.55); Common Scab records are the warmest (34.59 °C) with below-average soil moisture, matching the dry, higher-pH conditions described in [32]; Early Blight records carry the highest mean humidity (75.40%) and rainfall (3.89 mm); and Healthy records have by far the highest mean soil moisture (51.59%) and the lowest mean temperature (27.70 °C).

TABLE IV. MEAN ENVIRONMENTAL READINGS BY CLASS

Class	Temp. (°C)	Hum. (%)	Rain (mm)	Soil moist. (%)	Soil pH	Light (lux)
Anthrachnose	33.75	70.89	2.38	36.73	6.42	24,131
Common Scab	34.59	66.50	0.40	36.45	6.08	17,352
Early Blight	33.31	75.40	3.89	36.65	5.93	12,366
Fusarium Wilt	33.47	70.14	0.48	27.48	6.55	5,709
Healthy	27.70	66.11	3.21	51.59	6.14	12,937
Verticillium Wilt	28.66	63.63	0.21	34.94	5.95	12,370

Because the dataset is derived from geographically diverse Bangladeshi IoT deployments rather than from a single instrumented farm, it provides a realistic testbed for the low-cost, non-visual sensing scenario this study targets. It is, however, a single-country and single-source dataset, and Section VII returns to what that implies for generalisation.

IV. METHODOLOGY

The study follows a supervised tabular workflow that takes the labelled dataset of Section III and produces a trained six-class disease classifier together with a Shapley-based explanation of its predictions. The workflow comprises pre-processing, feature-target separation, imbalance handling, stratified splitting, model training, validation and explanation.

A. Data Pre-processing

Quality control confirmed that the working set contains no missing values, no duplicate rows, no infinite values and no constant features. Column names and categorical values were stripped of stray whitespace, and an empty-string check returned zero for every column. Feature ranges after cleaning are those reported in Table III: temperature 15.47–42.00 °C, humidity 42.00–88.99%, rainfall 0.00–50.00 mm, soil moisture 10.00–93.15%, soil pH 5.50–7.58 and light intensity 0–50,000 lux. As explained in Section III, the outliers identified by the interquartile-range test were retained, since they represent genuine rainfall events and diurnal light variation.

The two categorical attributes were one-hot encoded, producing four indicator columns and a ten-dimensional feature matrix X ; disease_label was retained as the target y . No scaling was applied, as neither tree ensemble is sensitive to monotone feature transformations.

B. Class-Imbalance Handling

Given the 9.19:1 ratio between the Healthy majority and the Common Scab minority, imbalance was addressed at two levels. First, both the train-test split and every cross-

validation fold were stratified on the disease label, preserving class proportions in each subset. Second, both classifiers were trained with balanced class weights—class_weight = "balanced" for Random Forest, and per-sample weights from compute_sample_weight(class_weight = "balanced") for XGBoost—so that minority classes contribute proportionally more to the training objective. The resulting sample weights ranged from 0.3403 to 3.1452 with a mean of 1.0.

C. Train-Test Split

An 80:20 stratified split with random_state = 42 produced 1,321 training and 331 test samples. The maximum class-proportion difference between the training and test partitions was 0.18 percentage points, and the two partitions were verified to share no observations. The test set was withheld from all training, model selection and cross-validation, and was evaluated exactly once.

D. Model Development

1) Random Forest

Random Forest was selected as the baseline for its robustness on tabular, mixed-type data. It grows an ensemble of decision trees on bootstrap resamples, choosing a random feature subset at each node. Each split minimises the weighted Gini impurity at node S ,

$$Gini(S) = 1 - \sum_k p_k^2,$$

where K is the number of disease classes and p_k the proportion of samples of class k at S . The ensemble prediction over T trees is obtained by majority vote,

$$fRF(x) = \arg \max_k (1/T) \sum_t \mathbb{I}(h_t(x) = k),$$

with h_t the prediction of tree t and $\mathbb{I}(\cdot)$ the indicator function. The configuration is given in Table V. Out-of-bag evaluation on the training set returned a score of 0.9326, an internal generalisation estimate obtained before the test set was used.

TABLE V. RANDOM FOREST HYPERPARAMETERS

Hyperparameter	Value
Number of trees (n_estimators)	500
Split criterion	Gini impurity
Max features per split	sqrt(d)
Class weighting	Balanced
Out-of-bag evaluation	Enabled
Random state	42

2) XGBoost

The second classifier is XGBoost, an additive model that fits trees sequentially so that each new tree corrects the residual error of the current ensemble. After T boosting rounds the model output for input x is

$$F(x) = \sum_t \eta \cdot h_t(x),$$

where η is the learning rate and h_t the regression tree added at round t . Each round selects h_t to minimise a regularised objective,

$$L^{(t)} = \sum_i \omega_i \cdot \ell(y_i, \hat{y}_i^{(t-1)} + h_t(x_i)) + \Omega(h_t),$$

in which ω_i is the balanced sample weight of instance i , ℓ the multiclass softmax cross-entropy loss and Ω a complexity penalty on the leaf weights of h_t . The configuration is given in Table VI. Both models were trained on the identical 1,321-sample, ten-feature training set.

TABLE VI. XGBOOST HYPERPARAMETERS

Hyperparameter	Value
Objective	multi:softprob (6 classes)
Number of trees (n_estimators)	500
Learning rate	0.05
Max tree depth	6
Subsample ratio	0.8
Column sample by tree	0.8
Tree construction method	Histogram-based
Class imbalance handling	Balanced sample weights
Random state	42

E. Evaluation Metrics

Both models were evaluated on the untouched 331-sample test set using accuracy, macro-averaged and weighted-averaged precision, recall and F1-score, per-class precision, recall, F1-score and support, and the full confusion matrix. Macro F1 was designated the primary selection metric: it weights all six classes equally and is therefore sensitive to minority-class behavior, whereas accuracy and weighted averages are dominated by the Healthy class, which alone accounts for 48.94% of the test set.

F. Cross-Validation and Statistical Testing

To establish that the single-split result is not an artefact of a favourable partition, both models were re-evaluated under repeated stratified K-fold cross-validation with five folds and five repeats, giving 25 independent train/validation runs. Cross-validation was confined to the training set; the held-out test set was not touched. Each run recorded training macro F1, validation macro F1 and validation accuracy. The train-validation macro F1 gap serves as an overfitting indicator, and a Wilcoxon signed-rank test over the 25 paired validation scores tests whether the difference between the two models is systematic. The Wilcoxon test was preferred over a paired t-test because fold scores are neither independent nor reliably normal.

G. Explainable AI

SHAP was applied to the better-performing model using shap.TreeExplainer, which computes exact Shapley values for tree ensembles. Values were computed over the full test set, yielding a $331 \times 10 \times 6$ array, and used at three levels: global importance as the mean absolute SHAP value per feature across all samples and classes; class-specific importance as a separate beeswarm summary per disease class; and local explanation as a waterfall decomposition of a single prediction into the base rate plus signed feature contributions. To guard against reporting an artefact of one attribution method, the SHAP ranking was cross-checked against Random Forest impurity importance and XGBoost gain importance using Spearman rank correlation.

H. Deployment-Style Inference

Finally, a reusable prediction function was implemented that accepts a raw unseen sensor reading, applies the fitted one-hot encoder and column alignment used in training, and returns each model's predicted class together with the full probability distribution. This demonstrates that the pipeline can consume new readings from a low-cost deployment without retraining.

A. Overall Performance

Table VII reports held-out test performance for both models. XGBoost outperformed Random Forest on every reported metric, reaching 95.17% accuracy and a macro F1-score of 0.9132 against 92.45% and 0.8584. The accuracy gap is 2.72 percentage points, but the macro F1 gap is larger at 0.0548, and macro recall shows the widest separation of all: 0.9161 for XGBoost against 0.8358 for Random Forest. That the macro-averaged advantage exceeds the accuracy advantage indicates that the improvement is concentrated in the smaller classes rather than in the already-saturated Healthy class. Fig. 4 presents the comparison graphically.

TABLE VII. HELD-OUT TEST-SET PERFORMANCE (N = 331)

Metric	Random Forest	XGBoost
Accuracy	92.45%	95.17%
Macro precision	0.8941	0.9113
Macro recall	0.8358	0.9161
Macro F1-score	0.8584	0.9132
Weighted precision	0.9234	0.9528
Weighted recall	0.9245	0.9517
Weighted F1-score	0.9216	0.9521

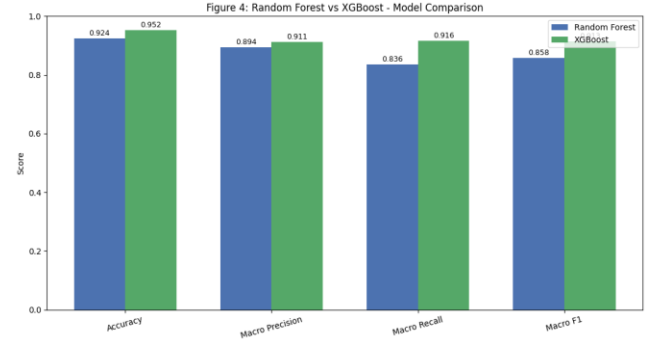


Fig. 4. Metric-by-metric comparison of Random Forest and XGBoost on the held-out test set. The separation is widest for macro recall and macro F1-score.

B. Per-Class Performance

Tables VIII and IX give the per-class classification reports, and the corresponding confusion matrices appear in Figs. 2 and 3. Both models are near saturation on the two largest classes: Healthy reaches an F1-score of 0.9725 under Random Forest and 0.9907 under XGBoost, and Fusarium Wilt reaches 0.9434 and 0.9709 respectively. The models diverge sharply on the minority classes. Random Forest recovers only 11 of 18 Anthracnose cases (recall 0.6111) and 12 of 18 Common Scab cases (recall 0.6667), whereas XGBoost recovers 16 and 15 respectively, lifting recall to 0.8889 and 0.8333. The largest single improvement is on Anthracnose, where F1 rises from 0.7333 to 0.9143, driven almost entirely by recall.

TABLE VIII. RANDOM FOREST PER-CLASS PERFORMANCE

Disease class	Precision	Recall	F1-score	Support
Anthracnose	0.9167	0.6111	0.7333	18
Common Scab	0.8000	0.6667	0.7273	18
Early Blight	0.9000	0.8571	0.8780	21
Fusarium Wilt	0.9091	0.9804	0.9434	51
Healthy	0.9636	0.9815	0.9725	162
Verticillium Wilt	0.8750	0.9180	0.8960	61

V. RESULTS

TABLE IX. XGBOOST PER-CLASS PERFORMANCE

Disease class	Precision	Recall	F1-score	Support
Anthracoese	0.9412	0.8889	0.9143	18
Common Scab	0.7500	0.8333	0.7895	18
Early Blight	0.9048	0.9048	0.9048	21
Fusarium Wilt	0.9615	0.9804	0.9709	51
Healthy	0.9938	0.9877	0.9907	162
Verticillium Wilt	0.9167	0.9016	0.9091	61

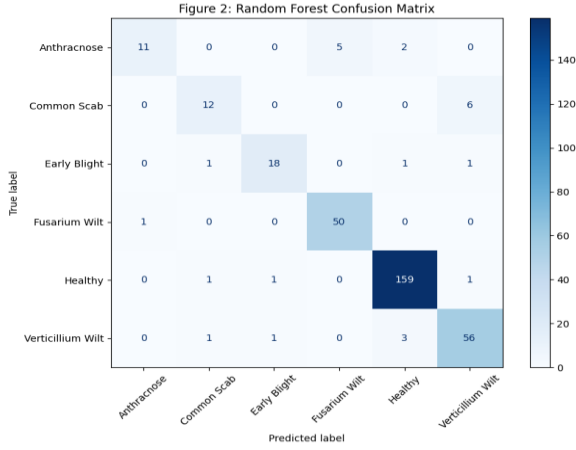


Fig. 2. Random Forest confusion matrix on the held-out test set. Missed Anthracnose cases are absorbed mainly by Fusarium Wilt, and missed Common Scab cases by Verticillium Wilt.

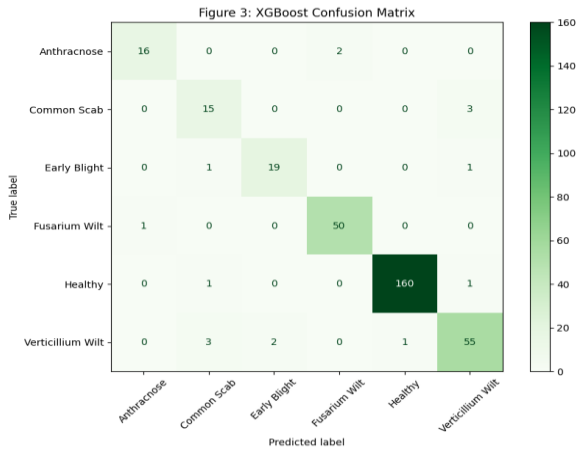


Fig. 3. XGBoost confusion matrix on the held-out test set. Minority-class recall improves substantially, though Common Scab remains the weakest class.

The single trade-off runs against XGBoost: its precision on Common Scab falls to 0.7500 from Random Forest’s 0.8000, because raising recall on that class introduces three additional false positives drawn from Verticillium Wilt. For an early-warning application this is an acceptable exchange, since the cost of a missed infection exceeds that of an unnecessary inspection. Common Scab nonetheless remains the weakest class for both models, at F1 = 0.7273 and 0.7895 respectively, so balanced class weighting has narrowed the imbalance penalty without eliminating it.

The confusion matrices show that the residual errors are structured rather than random. Under Random Forest, five of the seven missed Anthracnose cases are assigned to Fusarium Wilt and six of the six missed Common Scab cases to Verticillium Wilt. Table IV explains why: Anthracnose and Fusarium Wilt have closely comparable mean temperature (33.75 °C against 33.47 °C) and humidity (70.89% against

70.14%), so the classes overlap in the measured feature space. The errors are a consequence of genuine environmental similarity, not of model instability.

C. Cross-Validation and Statistical Significance

Table X summarises the 25-run repeated stratified cross-validation. XGBoost achieved a mean validation macro F1 of 0.8884 ± 0.0252 against 0.8526 ± 0.0341 for Random Forest, and the lower standard deviation indicates the advantage is also more stable across partitions. A Wilcoxon signed-rank test over the 25 paired validation scores returned $W = 12.0$ with $p < 0.001$, so the difference is statistically significant at $\alpha = 0.05$ and is not attributable to the choice of partition. It is worth noting that the same test applied to a single five-fold run ($n = 5$) returned $p = 0.1875$ and would not have reached significance; the repeated design was necessary to obtain adequate statistical power.

TABLE X. REPEATED STRATIFIED 5-FOLD CROSS-VALIDATION (25 RUNS)

Measure	Random Forest	XGBoost
Train macro F1 (mean $\pm$ std)	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
Validation macro F1 (mean $\pm$ std)	0.8526 $\pm$ 0.0341	0.8884 $\pm$ 0.0252
Validation accuracy (mean $\pm$ std)	0.9220 $\pm$ 0.0164	0.9397 $\pm$ 0.0136
Train-validation gap	0.1474	0.1116
Held-out test macro F1	0.8584	0.9132
Held-out – CV mean	0.0058	0.0248

Two further observations follow from Table X. First, both models reached a training macro F1 of exactly 1.0000, giving train-validation gaps of 0.1474 and 0.1116. Both exceed the 0.08 threshold conventionally treated as indicating high overfitting risk, which is expected for 500-tree ensembles fitted without early stopping or depth tuning on a dataset of this size. Second, and offsetting this, held-out test performance tracked the cross-validation mean closely for both models, with absolute differences of 0.0058 and 0.0248. The held-out result is therefore consistent with the cross-validated estimate rather than an optimistic outlier, even though the absolute level of both is inflated by memorisation of the training folds.

D. Explainability

SHAP was applied to XGBoost as the selected model. Table XI reports global importance as the mean absolute SHAP value per feature. Soil moisture is the strongest contributor at 1.3740, followed by the potato crop indicator at 1.0850 and temperature at 1.0094, with humidity fourth at 0.8027. The two season indicators contribute least, at 0.0626 and 0.0200, which is consistent with risk being governed by instantaneous micro-climate readings rather than by the coarse summer/winter label. Fig. 5 shows the corresponding beeswarm distribution, which resolves the direction as well as the magnitude of each contribution, and Fig. 6 presents the same ranking as a bar chart.

TABLE XI. GLOBAL SHAP FEATURE IMPORTANCE (XGBOOST, TEST SET)

Rank	Feature	Mean SHAP
1	soil_moisture	1.3740
2	crop_type = Potato	1.0850
3	temperature	1.0094
4	humidity	0.8027

Rank	Feature	Mean SHAP
5	light_intensity	0.5157
6	rainfall	0.4861
7	soil_ph	0.4821
8	crop_type = Tomato	0.2522
9	season = Summer	0.0626
10	season = Winter	0.0200

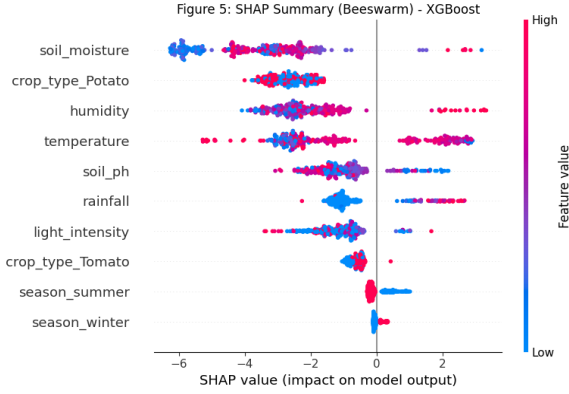


Fig. 5. SHAP beeswarm summary for XGBoost on the test set, showing the distribution and direction of each feature’s contribution.

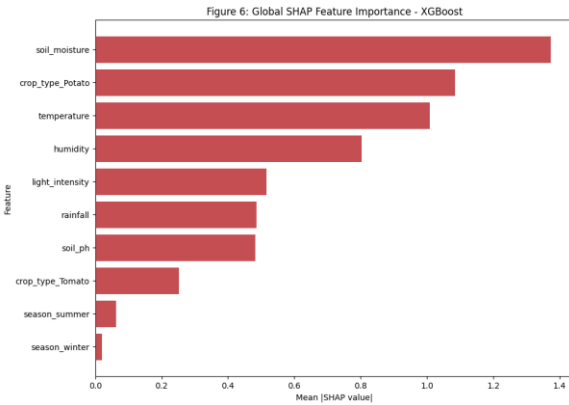


Fig. 6. Global SHAP feature importance for XGBoost, computed as the mean absolute SHAP value across all 331 test samples and all six classes.

The cross-check against the models’ built-in importance measures, reported in Table XII, produced the most consequential result of this section. SHAP and Random Forest impurity importance agree strongly, with Spearman $\rho = 0.8909$ ($p = 0.0005$): both rank soil moisture first, temperature second or third, and humidity in the top four. XGBoost’s own gain-based importance disagrees with both, at $\rho = -0.2848$ ($p = 0.4250$) against SHAP and $\rho = -0.4545$ ($p = 0.1869$) against Random Forest. Gain importance places the categorical indicators at the top—crop_type = Tomato at 0.2367 and season = Winter at 0.2294—and relegates soil moisture to 0.0484, below rainfall.

TABLE XII. FEATURE IMPORTANCE BY METHOD (NORMALISED)

Feature	RF (MDI)	XGB (gain)	SHAP
soil_moisture	0.2207	0.0484	0.2256
temperature	0.1660	0.0379	0.1657
humidity	0.1542	0.0323	0.1318
light_intensity	0.1041	0.0211	0.0847
crop_type = Potato	0.0889	0.1789	0.1782
soil_ph	0.0878	0.0208	0.0792
crop_type = Tomato	0.0700	0.2367	0.0414

Feature	RF (MDI)	XGB (gain)	SHAP
rainfall	0.0673	0.0403	0.0798
season = Summer	0.0206	0.1542	0.0103
season = Winter	0.0205	0.2294	0.0033

The disagreement has a known cause. Gain importance accumulates the loss reduction achieved at each split, which favours low-cardinality features that partition the data cleanly in early, high-gain splits; a binary crop indicator does exactly this. SHAP, by contrast, measures each feature’s marginal contribution to the predictions actually made on the test set. Reporting gain importance alone would have supported the conclusion that crop identity and season dominate disease risk—a conclusion that is agronomically implausible and, more seriously for a deployed system, would have implied that the real-time sensor stream contributes little. Because no feature appears in the top three of all three rankings, the SHAP and impurity consensus is adopted as the primary interpretive result.

E. Local Explanation and Deployment Behaviour

Global importance alone does not tell a grower why a particular warning was issued, so a local explanation was generated for an unseen reading: tomato in winter at 27.5 °C, 85.0% humidity, 12.4 mm rainfall, 65.0% soil moisture, pH 6.1 and 32,000 lux. XGBoost assigned Early Blight a probability of 43.61%, ahead of Anthracnose at 32.11% and Healthy at 23.27%. Table XIII decomposes the prediction and Fig. 7 presents the corresponding waterfall plot.

TABLE XIII. LOCAL SHAP CONTRIBUTIONS FOR THE EXAMPLE READING (PREDICTED CLASS: EARLY BLIGHT)

Feature	Value	SHAP value	Effect on risk
rainfall	12.4	+1.4162	Increases
temperature	27.5	−1.1129	Decreases
crop_type = Potato	0.0	−1.0637	Decreases
humidity	85.0	+1.0210	Increases
soil_moisture	65.0	+0.6295	Increases
light_intensity	32,000	−0.2316	Decreases

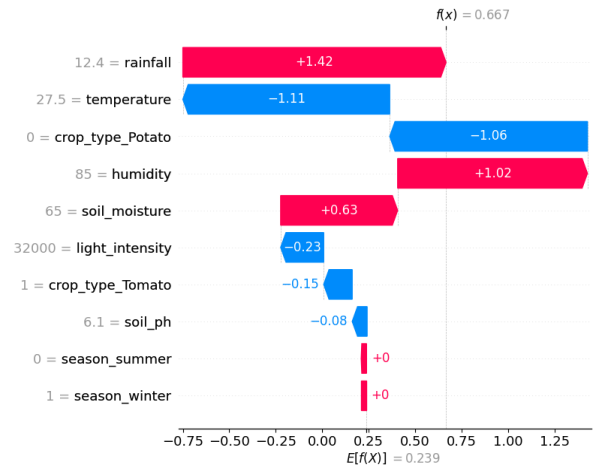


Fig. 7. SHAP waterfall decomposition for the example reading, expressing the Early Blight prediction as the base rate $E[f(X)] = 0.239$ plus the signed contribution of each feature, giving $f(x) = 0.667$.

The decomposition is agronomically legible. Rainfall of 12.4 mm is the strongest single driver at +1.4162, reinforced by humidity at 85.0% (+1.0210) and soil moisture at 65.0% (+0.6295); the elevated-moisture conditions that favour Early Blight are precisely what the model identified. The temperature reading of 27.5 °C pushes in the opposite

direction (-1.1129), since it falls below the $33.31\text{ }^{\circ}\text{C}$ mean of the Early Blight class in Table IV, which explains why the model returned 43.61% rather than near-certainty. A warning of this kind is actionable in a way a bare class label is not: it directs attention to surface wetness and drainage rather than to temperature.

The deployment function was additionally exercised on two further unseen readings. Both models agreed on the predicted class in each case, with XGBoost markedly more confident—97.36% against Random Forest’s 71.00% on the first, and 99.16% against 86.60% on the second. Agreement between two independently trained models is a useful practical signal, though the confidence gap also reflects the sharper probability distributions typical of boosted ensembles and should not be read as calibrated certainty.

VI. DISCUSSION

Finding 1: The cheap sensors aren’t just riding on correlation they’re actually picking up something real. When we used low-cost, consumer-grade sensor readings to separate six disease outcomes (no imagery involved), the macro-F1 scores came out strong enough to show these environmental variables carry genuine predictive signal. And because these readings are captured before any symptoms show up, they give growers something image-based systems simply can’t: an early warning. No matter how accurate a camera-based model like [10] or [11] gets, it still needs visible symptoms to work by definition, it can’t catch what hasn’t appeared yet. Finding 2: XGBoost beating Random Forest wasn’t a fluke. It held up across 25 folds, the gap was statistically significant, and it showed up exactly where it counts. Just balancing the class weights didn’t fix things Random Forest still missed about a third of Anthracnose and Common Scab cases even with `class_weight="balanced"` turned on. XGBoost’s boosting approach, which reweights the cases it gets wrong as it goes, recovered most of those missed infections. That matters because a missed infection keeps costing you, while a false alarm just costs one extra inspection so macro-F1 was the right metric to optimize for. Worth noting: a lazy “just predict healthy every time” baseline only scores 48.94% here, so the improvement is real, not just an artifact of an easy metric.

Finding 3: Flip the importance method, and the story completely changes. If we’d only looked at XGBoost’s built-in feature importance, we’d have concluded crop type and season drive disease risk. SHAP tells a different story entirely soil moisture, temperature, and humidity dominate, while season barely matters. The two rankings actually disagree ($\rho = -0.2848$), which is a real problem if you’re only using one attribution method. This is exactly why we’d argue agricultural XAI work should cross-check across at least two importance methods before drawing conclusions growers might act on. Here, SHAP and impurity importance actually agree strongly ($\rho = 0.8909$), which is what makes the soil moisture/temperature/humidity finding trustworthy — and it lines up with what plant pathologists already know about wetness and temperature needing to co-occur for infection. Though it’s fair to note the labels themselves were built from thresholds on these same variables, so some of that agreement is expected.

VII. LIMITATIONS

Several constraints should be considered when interpreting the results of this study. First, disease labels were derived from published environmental threshold conditions [28]–[32] rather than directly observed or laboratory-confirmed cases, creating a potential circularity risk. Therefore, the reported accuracy reflects the model’s ability to learn a consistent mapping from the available data rather than validated field diagnostic performance. Second, both models achieved a training macro F1-score of 1.0000, with train-validation gaps of 0.1474 and 0.1116, indicating a notable risk of overfitting and potentially optimistic absolute performance. Third, the study uses a single Bangladeshi IoT dataset [27], limiting generalisation across locations, climates, soil conditions, crops, and sensor hardware. The dataset also covers only potato and tomato and six disease classes, while minority classes have relatively few test samples, reducing the reliability of their individual performance estimates. Furthermore, the absence of temporal modelling prevents the framework from capturing the cumulative effects of environmental conditions on disease development [1], [4]. The explainability analysis is also asymmetric because SHAP was applied only to XGBoost, while other feature-importance measures were used for comparison. Finally, the framework has not been evaluated in real-world field conditions, where sensor noise, missing readings, calibration drift, and inference latency may affect performance. Future work should therefore focus on independently validated labels, multi-location datasets, temporal modelling, and real-world IoT deployment.

VIII. CONCLUSION AND FUTURE WORK

This study examined whether crop-disease risk can be predicted from low-cost IoT micro-climate readings alone, and whether such predictions can be explained well enough to be acted upon. Using a held-out test set on a 1,652-record data set containing potato and tomato across six outcome classes, XGBoost outperformed Random Forest, with 95.17% accuracy and 0.9132 Macro F1, a difference that consistently supported 25 repeated cross validation runs and demonstrated a significant difference under a Wilcoxon signed-rank test. SHAP analysis concluded that soil moisture, crop type and temperature were the most influential contributing factors, which aligns with variable importance based on Random Forest impurity but is inconsistent with XGBoost gain importance – a result which would have led to a different conclusion on the agronomic factors to be considered, had only one attribution method been relied upon. Local explanations decomposed individual predictions into signed, physically interpretable contributions, showing that the framework can state which measured condition drove a warning.

These results support the feasibility of a lightweight, interpretable, sensor-driven early-warning aid for urban and balcony cultivation. Two qualifications are essential. These labels were based on published rules for thresholds and there is a fairly large train/validation gap in both models. It is therefore important to consider the framework as a screening tool and to be used in conjunction with visually based methods of inspection and expert diagnosis. Future work

follows directly from these limitations. The priority is field validation: collecting paired sensor readings and agronomist-confirmed diagnoses through balcony trials or extension-service partnerships would replace rule-derived labels and resolve the circularity concern. Second, systematic hyperparameter optimization with early stopping, together with resampling strategies such as SMOTE or explicit cost-sensitive losses, should narrow the train-validation gap and improve minority-class recall. Third, extending the feature representation from instantaneous snapshots to environmental sequences would allow recurrent or temporal-attention models to capture the duration effects that govern infection [1], [4]. Fourth, multi-modal fusion with image-based leaf classification [3], [10], [11] would allow a single system to issue a pre-symptomatic warning and subsequently confirm it once symptoms appear. Finally, broadening coverage to additional crops, disease classes and climatic regions, and evaluating on-device inference under realistic sensor noise, would establish whether the approach transfers beyond the conditions studied here.

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